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BATCH - 16 BTECH CSE

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Algorithm

Right-Aligned Increasing Triangle

1. Start.
2. Input the number of rows n .
3. Set $i = 1$.
4. Repeat for every row from 1 to n .
5. Print $n-i$ spaces.
6. Print i stars.
7. Move to the next line.
8. Stop.

Right-Aligned Increasing Number Triangle

- a. Start.
- b. Input the number of rows n .
- c. Set $i = 1$.
- d. Repeat for every row from 1 to n .
- e. Print $n-i$ spaces.
- f. Print numbers from 1 to i .
- g. Move to the next line.

h. Stop.

Right-Aligned Decreasing Triangle

- a. Start.
- b. Input the number of rows n .
- c. Set $i = 1$.
- d. Repeat for every row from 1 to n .
- e. Print $i-1$ spaces.
- f. Print stars from i to n .
- g. Move to the next line.
- h. Stop.

C program

```
#include <stdio.h>
```

```
// Function 1: Right-aligned increasing triangle
```

```
void increasingStar()
```

```
{
```

```
    int n, i, j;
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &n);
```

```
    for (i = 1; i <= n; i++)
```

```
{  
    // Print spaces  
    for (j = 1; j <= n - i; j++)  
        printf(" ");  
  
    // Print stars  
    for (j = 1; j <= i; j++)  
        printf("* ");  
  
    printf("\n");  
}  
}
```

// Function 2: Right-aligned increasing number triangle

```
void increasingNumber()  
{  
    int n, i, j;  
  
    printf("Enter number of rows: ");  
    scanf("%d", &n);  
  
    for (i = 1; i <= n; i++)  
    {
```

```
// Print spaces
```

```
for (j = 1; j <= n - i; j++)
```

```
    printf(" ");
```

```
// Print numbers
```

```
for (j = 1; j <= i; j++)
```

```
    printf("%d ", j);
```

```
printf("\n");
```

```
}
```

```
}
```

```
// Function 3: Right-aligned decreasing triangle
```

```
void decreasingStar()
```

```
{
```

```
    int n, i, j;
```

```
    printf("Enter number of rows: ");
```

```
    scanf("%d", &n);
```

```
    for (i = 1; i <= n; i++)
```

```
    {
```

```
        // Print spaces
```

```
for (j = 1; j <= i - 1; j++)
```

```
    printf(" ");
```

```
// Print stars
```

```
for (j = 1; j <= n; j++)
```

```
    printf("* ");
```

```
printf("\n");
```

```
}
```

```
}
```

```
int main()
```

```
{
```

```
    int choice;
```

```
printf("\n===== MENU =====\n");
```

```
printf("1. Right-aligned increasing triangle\n");
```

```
printf("2. Right-aligned increasing number triangle\n");
```

```
printf("3. Right-aligned decreasing triangle\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch (choice)
```

```
{  
  
    case 1:  
        increasingStar();  
        break;  
  
    case 2:  
        increasingNumber();  
        break;  
  
    case 3:  
        decreasingStar();  
        break;  
  
    default:  
        printf("Invalid choice!\n");  
}  
  
return 0;  
}
```

Session Actions Edit View Help

```
(kali@kali)-[~/Desktop]
$ gcc sample21.c -o sample21
```

```
(kali@kali)-[~/Desktop]
$ ./sample21
```

```
==== MENU ====
```

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle

Enter your choice: 1

Enter number of rows: 6

```
      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * * *
```

```
(kali@kali)-[~/Desktop]
$ gcc sample21.c -o sample21
```

```
(kali@kali)-[~/Desktop]
$ ./sample21
```

```
==== MENU ====
```

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle

Enter your choice: 2

Enter number of rows: 4

```
      1
     1 2
    1 2 3
   1 2 3 4
```

```
(kali@kali)-[~/Desktop]
$ ./sample21
```

```
==== MENU ====
```

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle

Enter your choice: 3

Enter number of rows: 7

```
* * * * * * *
 * * * * * *
  * * * * *
   * * * *
    * * *
     * *
      *
       *
```

```
(kali@kali)-[~/Desktop]
$
```